

Philosophy of Science
Prof. James W. McAllister

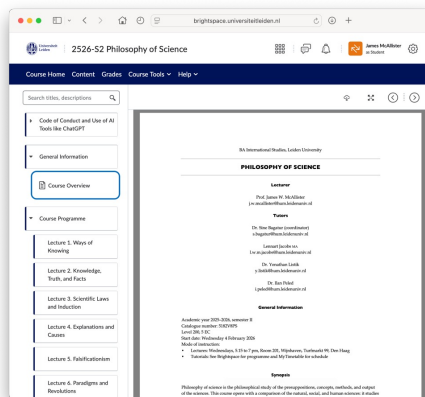
Lecture 1 Ways of Knowing

Welcome!

- Introducing...
 - Your lecturer
 - This course “Philosophy of Science”
- Your tutorial team
 - Sine Bagatur (coordinator)
 - Lennart Jacobs
 - Yonathan Listik
 - Ilan Peled

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Full information on Brightspace



- Complete programme
- Powerpoint presentations
- Links to readings
- Livestreams
- Videorecordings
- Tutorial assignments

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Why “Philosophy of science”?

- This module offers you two things
 - To complement your thematic courses
- Metacognition—reflection on cognition
 - How do disciplines conceptualise the world?
 - What does it mean to have knowledge?
 - How do scientists reason?
- Tools to solve problems as yet unknown
 - Value of abstract conceptual frameworks
 - Example: generative artificial intelligence

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Structure of the course

- This module is composed of:
 - Eleven lectures covering five themes
 - Including a final review and question period
- Tutorials—see MyTimetable for schedule

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I. The basics

1. Ways of knowing ← *Today*
 - Variety of the natural, social, and human sciences
 - Nomothetic and idiographic approaches
2. Knowledge, truth, and facts
 - Propositional knowledge: “I know that p ”
 - Knowledge as justified true belief
 - Truth as correspondence with facts

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II. Some scientific devices

3. Scientific laws and induction
 - Nomothetic approach—in social and human sciences too
 - Induction vs. deduction
 - David Hume’s problem of induction and a pragmatic response
4. Explanations and causes
 - Classic models of explanation, e.g., deductive-nomological model
 - Requirement to track causal links

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III. Models of scientific practice

5. Falsificationism
 - Karl R. Popper: science as trial and error
 - Demarcation of science from non-science
6. Paradigms and revolutions
 - Thomas S. Kuhn: theory-ladenness of observation
 - Normal science: paradigms and puzzles
 - Scientific revolutions, incommensurability

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IV. Methodological tool-box

7. Analogies and models
 - Structure of analogies: mappings and relations
 - Models as idealised depictions of reality
8. Quantitative and qualitative approaches
 - Reasoning between causes and effects
 - Hypothesis formulation and hypothesis test
 - Case study methodology

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V. Rationality and culture

9. Rationality
 - Rational choice theory and its limitations
 - Bounded rationality, role of emotion
10. Interpretation and understanding
 - Idiographic approach: Wilhelm Dilthey
11. Absolutism and relativism
 - Does validity depend on background?
 - Relativism about rationality
12. Review and question period

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LOST YOUR MAIL? SCAN THE CODE

NATIONALE STUDENTEN ENQUETE

NSE: het landelijk toezichtsbureau

The graphic features a smiling young woman with blonde hair wearing white headphones, holding a white smartphone and a brown coffee cup. The background is yellow with a white abstract shape behind her. A QR code is located in the bottom right corner of the graphic area.

Overview of today's lecture

- Three groups of sciences
 - Natural, human, and social sciences
 - Their characteristics, strengths, and blind spots
- Distinction between two approaches to describe the world
 - Nomothetic and idiographic approaches

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Motivation of today's lecture

- In International studies...
 - You are required to deal with knowledge from many disciplines
 - Human sciences, social sciences—and even natural sciences
- It is essential that you know what kinds of knowledge these disciplines produce
 - Emphasis on multidisciplinary

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Multidisciplinary: introduction

- Integration of knowledge and methods...
 - from multiple academic disciplines
 - to address complex societal problems
- Characteristics
 - Collaboration among different fields
 - Diversity of perspectives
 - Greater problem-solving capacity
- Relevant to...
 - Climate crisis, migration, conflict resolution

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Multidisciplinary: example

- Europe-wide nitrogen crisis
 - Dutch *stikstofcrisis*, since 2019
 - Requires a multidisciplinary approach
- Chemistry and ecology
 - Ammonia and nitrogen oxides in soil, water
- Political science, economics, geography
 - Government policies, business viability
- Humanities
 - Senses of history, belonging, recognition

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Reading for this lecture

- Jerome Kagan, *The Three Cultures: Natural Sciences, Social Sciences, and the Humanities in the 21st Century*
 - Cambridge University Press, 2009
 - Read chapter 1, "Characterizing the Three Cultures", pp. 1–50
 - Available on Brightspace
- Purpose of the readings

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Ways of knowing

- Broad set of practices including:
 - Epistemologies of different cultures
 - Knowledge incorporated in artworks, such as novels and poetry
 - People's everyday and tacit knowledge
 - Common sense
- Our focus here:
 - Mainstream Western sciences and scholarly disciplines and their diversity

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Three groups of sciences

- Natural sciences
 - Physics, chemistry, biology, astronomy...
- Human sciences
 - History, art history, linguistics, literary studies, philosophy, religious studies...
- Social sciences
 - Sociology, political science, economics, psychology, anthropology...

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Natural sciences 1

- Ancient intellectual endeavours
 - Astronomy arose in Mesopotamia, 1200 BCE
- Modern natural sciences
 - Developed partly from Chinese, Indian, and Islamic sources in Middle Ages and Renaissance
- "Scientific revolution"
 - Europe, 1550–1700

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Natural sciences 2

- Physical universe is uniform and simple
- Natural sciences focus on universals and regularities
 - Example: phenomenon of free fall
 - Less interest in concrete historical particulars, such as the fall of this stone
- Standard techniques of theorising:
 - Mathematisation, abstraction, idealisation
 - Yield knowledge in concise, powerful forms

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Natural sciences 3

- Example: laws of nature
 - Mathematical equations among physical quantities
 - Isaac Newton's law of gravitation, 1687:
$$F = Gm_1m_2/r^2$$
- Laws as paradigms of knowledge
 - Taken to represent the highest grade of scientific knowledge
 - Even by many outside the natural sciences

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Human sciences 1

- Liberal arts in ancient and medieval education
 - Grammar, logic, rhetoric
 - "Ways of doing" rather than of knowing
- Transformation into subjects of study
 - Renaissance humanism, 16th century
- Formalisation of human sciences
 - Germany, 19th century
 - History, history of art, studies of language and literature, philosophy, religious studies

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Human sciences 2

- The world studied by the human sciences
 - Focuses on historical human actors
- Historical actors are creative
 - They originate acts, texts, artworks
 - Creation follows no rules—unpredictable and inexplicable
- Important methodological consequences...

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Human sciences 3

- Historical particularity
 - Every event and context is unique
 - We identify periods (e.g., Renaissance, Enlightenment), but then zoom in past these categories
- Mistrust of generalisation and idealisation
 - Highest form of knowledge is intimate knowledge of particulars
 - Little or no use for scientific laws

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Human sciences 4

- Main output: interpretations
 - Of acts, texts, artworks
 - Often embedded in theoretical frameworks
- Empathy, hermeneutics
 - We attempt to reconstruct the historical actor's world of experience and meanings
- Objectivity of interpretation
 - We test interpretations against the text or other material

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Some complications

- Both groups of disciplines are more diverse than I have described
- Styles of theorising in natural sciences
 - Example: mechanism thinking in biology
- Generalisation in human sciences
 - Linguistics, archaeology use some methods inspired by the natural sciences
 - Marxist historiography: belief in universal laws of historical development

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Question

- How can we...
 - conceptualise the differences between natural and human sciences?
 - analyse the diversity of the social sciences?
- Answer: using the concepts of nomothetic and idiographic approaches
 - Proposed by Wilhelm Windelband (1848–1915), German neo-Kantian philosopher, around 1894

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Nomothetic approach 1

- What does the nomothetic approach consist in?
 - Identifying regularities in the world
 - Formulating generalisations and laws to describe these regularities
 - Deriving explanations of observed outcomes from these generalisations and laws
- Typical of the natural sciences
 - But not unknown in human and social sciences—see lecture 3

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Nomothetic approach 2

- Strength: nomothetic approach can...
 - identify similarities and structures that underlie apparently diverse cases
 - yield sweeping, general knowledge
 - yield economical knowledge
- Weakness: nomothetic approach can...
 - erase the specificity of outcomes
 - be reductive, mechanistic, positivistic

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Idiographic approach 1

- What does the idiographic approach consist in?
 - Understanding the meaning of contingent, unique, and often subjective outcomes
- Typical of the human sciences
 - Windelband's term: *Geisteswissenschaften*, or "sciences of the spirit"

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Idiographic approach 2

- Strength: idiographic approach can...
 - reveal differences between apparently similar cases
 - yield detailed, context-sensitive knowledge
- Weakness: idiographic approach can...
 - be blind to general factors that constrain outcomes

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Review: nomothetic/idiographic

- Nomothetic approach: tendency to...
 - Generalise
 - Explain outcomes by appeal to general rules
- Idiographic approach: tendency to...
 - Specify
 - Understand the meaning of unique, contingent, and often subjective acts
- Tension between these approaches
 - Felt particularly in social sciences

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Social sciences 1

- Youngest group of disciplines
 - Originated in late-19th-century French and German debates on how to study societies
 - Auguste Comte (1798–1857), Emile Durkheim (1858–1917), Max Weber (1864–1920)
- Present-day disciplines:
 - Sociology, political science, economics, psychology, anthropology...

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Social sciences 2

- Social sciences feel the pull of both natural and human sciences
- Economics, demography
 - Largely nomothetic disciplines
 - Predominantly mathematical investigation of underlying phenomena
- Cultural anthropology, political theory
 - Largely idiographic disciplines
 - Produce interpretations and ascribe meanings

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Social sciences 3

- Diversity even within single disciplines
 - Example: psychology
- Nomothetic approach
 - Psychometric approach to personality
 - Categorises individuals in terms of underlying universal traits or dimensions
- Idiographic approach
 - Sigmund Freud's analysis of "kleine Hans", a child with a phobia of horses, 1909
 - Clinical notes running to 150 pages

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Recapitulation

- Characteristics of...
 - Natural, human, and social sciences
 - Nomothetic and idiographic approaches
- This overview gives you a basis to...
 - interpret what knowledge each discipline can produce
 - integrate the output of diverse disciplines

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